Volumes of Revolution

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.02

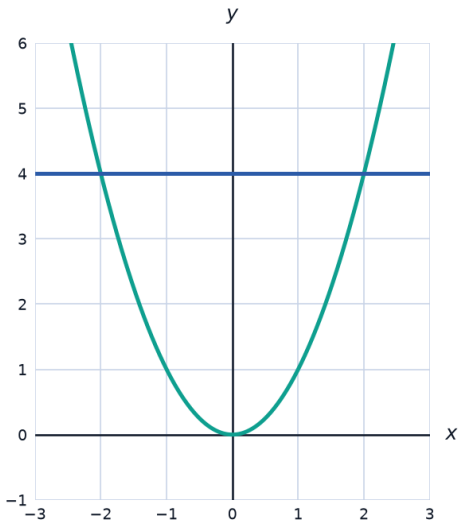
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The disk and washer methods**

Find the volume generated when the region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis.

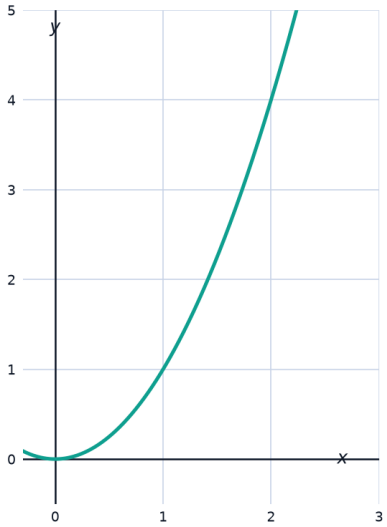
2 **[Foundational]** • **Target: The disk and washer methods**

The region between $y = x^2$ and $y = 4$ is revolved about the x-axis. Using the washer method, set up and evaluate the volume integral.



3 **[Foundational]** • **Target: The shell method**

The region under $y = x^2$ on $[0, 2]$ is revolved about the y-axis. Use the shell method to find the volume.



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4 [Foundational] • Target: The shell method

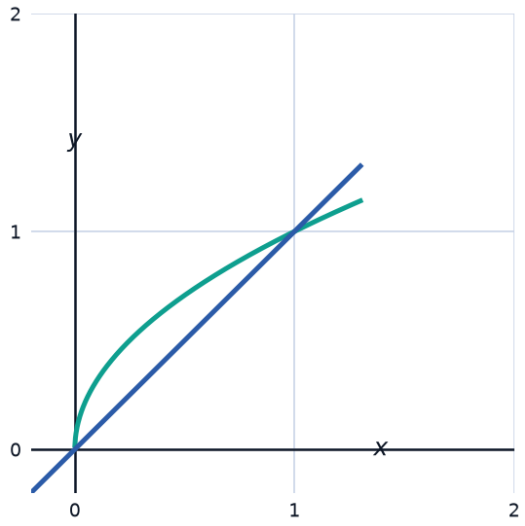
The region bounded by $y = x$, $y = 0$, and $x = 3$ is revolved about the y-axis. Find the volume using the shell method.

5 [Foundational] • Target: Comparing methods

The region under $y = x^2$ on $[0, 2]$ is revolved about the x-axis (not the y-axis). Find the volume using the disk method.

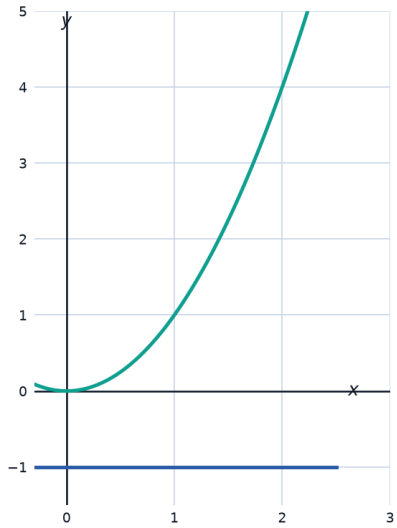
6 [Foundational] • Target: Comparing methods

The region bounded by $y = \sqrt{x}$ and $y = x$ (between their two intersection points) is revolved about the x-axis. Find the volume using the washer method.



7 [Foundational] • Target: Volumes about other axes

The region under $y = x^2$ on $[0, 2]$ is revolved about the line $y = -1$. Find the volume, using outer radius $x^2 + 1$.



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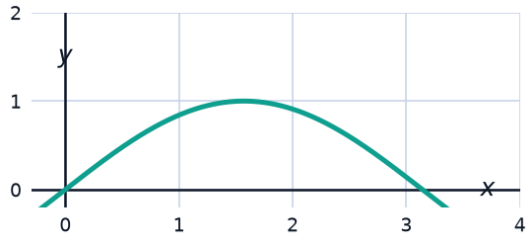
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Volumes about other axes**

The region under $y = x^2$ on $[0, 2]$ is revolved about the line $x = 3$ (using the shell method with radius $3 - x$).

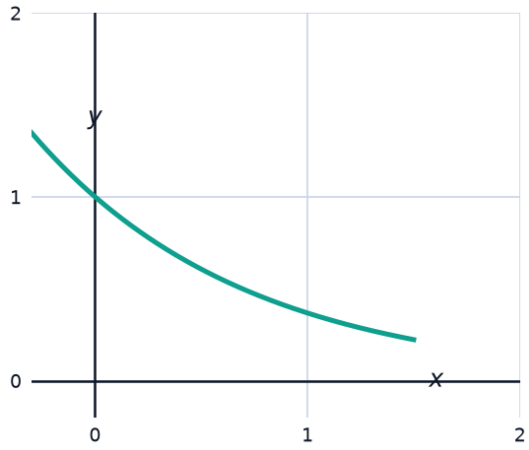
9 **[Intermediate]** • **Target: Volumes with trigonometric and exponential curves**

The region under $y = \sin x$ on $[0, \pi]$ is revolved about the x-axis. Find the volume, using $\sin^2 x = \frac{1 - \cos 2x}{2}$.



10 **[Intermediate]** • **Target: Volumes with trigonometric and exponential curves**

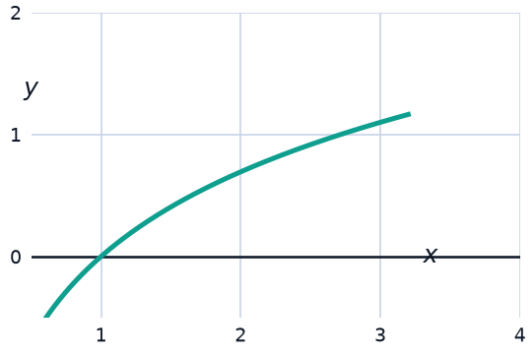
The region under $y = e^{-x}$ on $[0, 1]$ is revolved about the x-axis. Find the volume, to three decimal places.



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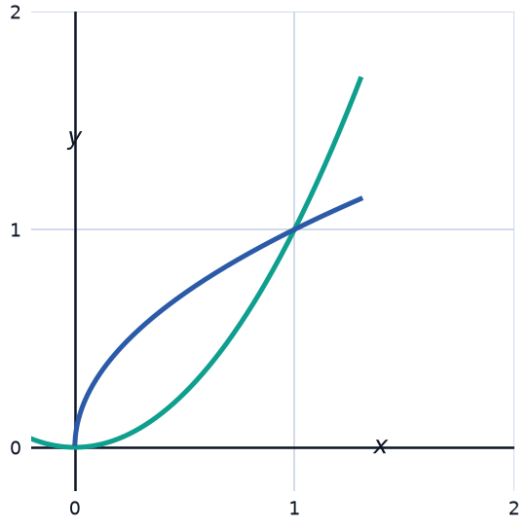
11 **[Intermediate]** • **Target: Setting up without evaluating**

Set up (do not evaluate) the volume integral for the region under $y = \ln x$ on $[1, e]$, revolved about the x-axis.



12 **[Intermediate]** • **Target: Setting up without evaluating**

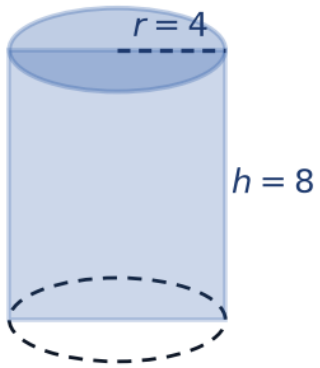
Set up (do not evaluate) the volume integral for the region between $y = x^2$ and $y = \sqrt{x}$ (on $[0, 1]$) revolved about the y-axis, using the shell method.



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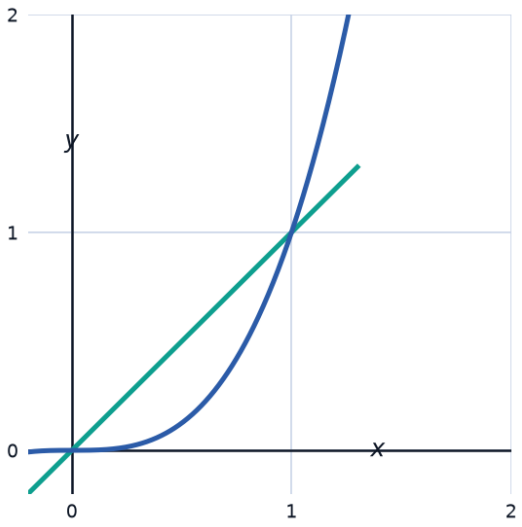
13 **[Intermediate]** • **Target: Visualizing the generating region**

The region under $y = \sqrt{x}$ on $[0, 4]$ (from an earlier disk-method problem) is shown. Sketching this region mentally revolved about the x-axis produces a solid — describe its shape, and confirm the volume 8π found earlier is reasonable by comparing to a cylinder of radius 2 and length 4.



14 **[Intermediate]** • **Target: Volume of a solid with a hole (washer method practice)**

The region between $y = x$ and $y = x^3$ on $[0, 1]$ is revolved about the x-axis. Find the volume using the washer method (outer radius x, inner radius x^3).

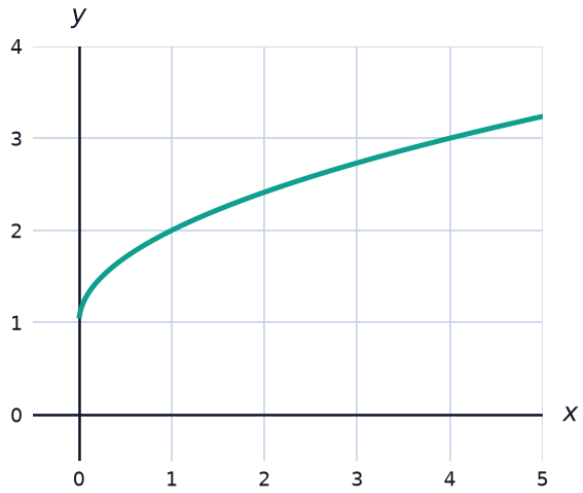


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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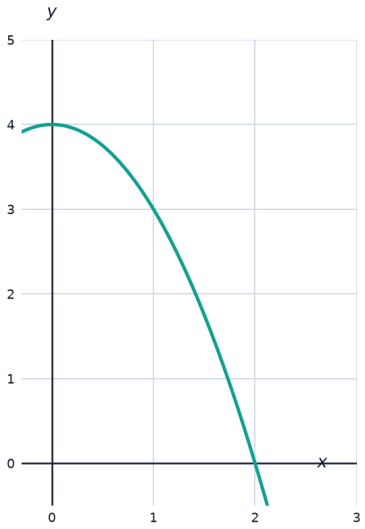
15 [Challenging] • Target: An applied volume problem • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

A vase is modeled by revolving the region under $y = \sqrt{x} + 1$ on $[0, 4]$ about the x-axis, where x is the height (cm) and y is the radius. Find the volume of the vase.



16 [Challenging] • Target: Volume with the shell method, revisited • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

The region bounded by $y = 4 - x^2$ and $y = 0$ (for $x \geq 0$) is revolved about the y-axis. Use the shell method to find the volume.



17 [Challenging] • Target: Visualizing the generating region • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Find the volume of the solid generated when the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is revolved 2π radians about the x-axis.

18 [Challenging] • Target: Volume of a solid with a hole (washer method practice) • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

(IB AA/AI HL) Given the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's numerical method with step size $h = 0.1$ to approximate $y(0.2)$. Compare your numerical result with the exact analytical solution.

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19 [Challenging] • Target: An applied volume problem • Cross-Lesson: 6.01 Related Rates

[Unit 7BC · Logistic Models & Euler's Method · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 [Challenging] • Target: Volume with the shell method, revisited • Cross-Lesson: 6.02 Optimization and Curve Sketching

[About A2 Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The disk and washer methods

Shell method for solid of revolution about y-axis: cylindrical shells have radius $=$

a. $f(x)$

b. $b - a$

c. a

d. x

22 [Foundational] • Target: The shell method

Volume of solid of revolution of $y = \cos x$ about x-axis from 0 to $\pi/2$:

a. $\pi^2/2$

b. $\pi/2$

c. $\pi^2/4$

d. $\pi/4$

23 [Foundational] • Target: Comparing methods

Volumes: for the same region rotated, which is larger: shell or washer?

a. Shell is always larger

b. They give the same volume

c. Washer is always larger

d. Depends on axis and region

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24 [Foundational] • Target: Volumes about other axes

Which gives volume by the disk method about the y-axis?

a. $\int_a^b f(y)\,dy$

b. $\pi\int_a^b [f(x)]^2\,dx$

c. $2\pi\int_a^b xf(x)\,dx$

d. $\pi\int_c^d [f(y)]^2\,dy$ (solve for x first)

25 [Foundational] • Target: Volumes with trigonometric and exponential curves

Volume of the solid of revolution of $y = \sqrt{9 - x^2}$ (semicircle) about x-axis:

a. 18π

b. $\frac{4}{3}\pi \cdot 27 = 36\pi$

c. 36π

d. $\frac{4}{3}\pi(3)^3 = 36\pi$

26 [Foundational] • Target: Setting up without evaluating

A region R is bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. A solid is formed by revolving this region R about the y-axis. Find the volume of this solid. \\ \\ \\ (Hint: Consider the shell method and use a substitution that relates to the derivative of the denominator of the integrand.)

a. $\pi\ln(10)$

b. $2\pi\ln(5)$

c. $\pi\ln(5)$

d. $2\pi\ln(10)$

27 [Foundational] • Target: Visualizing the generating region

Volume by shell about x-axis: shells have radius $=$

a. y

b. x

c. $f(x)$

d. $f(y)$

28 [Foundational] • Target: Volume of a solid with a hole (washer method practice)

The region under $y = \sin(x)$ for $0 \le x \le \pi$ is revolved about the x-axis. Find the volume.

a. $\frac{\pi}{2}$

b. π^2

c. $\frac{\pi^2}{2}$

d. $2\pi^2$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: An applied volume problem**

Volume by disc method rotating $y = \sqrt{x}$ about x-axis from 0 to 4:

a. 4π

b. 16π

c. 8π

d. 32π

30 **[Intermediate]** • **Target: Volume with the shell method, revisited**

Volume of the region $0 \le x \le 1$, $0 \le y \le e^x$ rotated about $y = -1$:

a. $\pi(e^2 - 2e + 1)/2$

b. $\pi(e^2 - 2e)/2$

c. $\pi(e^2 - 3)/2$

d. $2\pi(e - 2)$

31 **[Intermediate]** • **Target: The disk and washer methods**

Volume of the solid obtained rotating $y = \sin x$ $(0 \le x \le \pi)$ about the x-axis:

a. $\pi^2/4$

b. $\pi^2/2$

c. $\pi/2$

d. π

32 **[Intermediate]** • **Target: The shell method**

Pappus' theorem: solid of revolution of right triangle $(0, 0), (1, 0), (0, 1)$ about y-axis: $\bar{x} = 1/3, A = 1/2$:

a. $\pi/2$

b. $\pi/6$

c. $\pi/3$

d. $2\pi/3$

33 **[Intermediate]** • **Target: Comparing methods**

The region between $y = 2$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{19}{5}$

b. $\frac{19}{10}\pi$

c. $\frac{19}{5}\pi$

d. $\frac{38}{5}\pi$

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34 **[Intermediate]** • **Target: Volumes about other axes**

Find the volume of the solid inside the sphere $\rho = 2$ and above the cone $\phi = \frac{\pi}{4}$.

a. $2\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

b. $4\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

c. $\frac{16\pi}{3}$

d. $2\sqrt{2}\,\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

35 **[Intermediate]** • **Target: Volumes with trigonometric and exponential curves**

Volume rotating $y = 2$ about the x-axis from 0 to 3:

a. 4π

b. $4\pi \cdot 3 = 12\pi$

c. 6π

d. 12π

36 **[Intermediate]** • **Target: Setting up without evaluating**

Moment of inertia of spherical shell $x^2 + y^2 + z^2 = R^2$ about z-axis using surface integral:

a. $\frac{2}{5}MR^2$

b. $\frac{2}{3}MR^2$

c. MR^2

d. $\frac{1}{2}MR^2$

37 **[Intermediate]** • **Target: Visualizing the generating region**

Volume of spherical shell $1 \le \rho \le 2$:

a. 4π

b. $28\pi/3$

c. 8π

d. $4\pi/3$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Volume of a solid with a hole (washer method practice) • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

For a region symmetric about the y-axis, $\iint_D x\,dA =$

a. 0

b. -1

c. 1

d. $A(D)$

39 [Challenging] • Target: An applied volume problem • Cross-Lesson: 6.01 Related Rates

Volume of solid V using divergence theorem: $V = \frac{1}{3}\oiint_S \mathbf{r} \cdot d\mathbf{S}$ since $abla \cdot \mathbf{r} =$

a. 3

b. 1

c. 0

d. $\mathbf{r}$

40 [Challenging] • Target: Volume with the shell method, revisited • Cross-Lesson: 6.02 Optimization and Curve Sketching

Find the volume of the solid between the paraboloid $z = x^2 + y^2$ and the plane $z = 1$.

a. π

b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$

d. $\frac{3\pi}{2}$

41 [Challenging] • Target: The disk and washer methods • Cross-Lesson: 7.01 Integration by Substitution

Volume of region $0 \le z \le 1 - x^2 - y^2$ (positive):

a. $\pi/4$

b. $\pi/2$

c. π

d. $3\pi/4$

42 [Challenging] • Target: The shell method • Cross-Lesson: 7.02 Integration by Parts

Find the volume of the solid under $z = 3x + y$ above the unit square $[0, 1] \times [0, 1]$.

a. 2

b. 4

c. 1

d. 6

43 [Challenging] • Target: Comparing methods • Cross-Lesson: 8.01 Area Between Curves

Kelvin-Stokes theorem applied to $\mathbf{F} = (y, 0, 0)$ over disk $z = 0, x^2 + y^2 \le 1$:

a. 2π

b. π

c. 0

d. $-\pi$

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44 [Challenging] • Target: Volumes about other axes • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Volume by shells revolving $y = x^2$ about the y-axis from 0 to 1:

a. $\pi/2$

b. $\pi/4$

c. π

d. $2\pi/4$

45 [Challenging] • Target: Volumes with trigonometric and exponential curves • Cross-Lesson: 9.01 Separable Differential Equations

Volume of solid with square cross-sections perpendicular to x-axis, bounded by $y = \sqrt{x}$ and $x = 4$:

a. 4

b. 8

c. 16/3

d. 8/3

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